p5.js

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Shape

2D Primitives

arc() Draws an arc to the screen.	ellipse() Draws an ellipse (oval) to the screen.	circle() Draws a circle to the screen.	line() Draws a line (a direct path between two points) to the screen.	point() Draws a point to the screen.
quad() Draws a quadrilateral to the screen.	rect() Draws a rectangle.	square() Draws a square.	triangle() Draws a triangle.	ellipseMode() Sets how arc(), ellipse(), and rect() interpret their parameters.
noSmooth() Disables smoothing when drawing to the canvas.	rectMode() Sets how rect() interprets its parameters.	strokeCap() Sets the shape of the ends of lines.	strokeJoin() Sets the shape of the joins used for polylines, and the shape of the corners for rectangles.	strokeWeight() Sets the thickness of lines.

3D Primitives

beginContour() Begins adding a new hole within a shape.	box() Draws a rectangular prism.	ellipsoid() Draws an ellipsoid.	plane() Draws a plane.	sphere() Draws a sphere.
cone() Draws a cone.	cylinder() Draws a cylinder.	endContour() Ends adding a hole within a shape.	torus() Draws a torus.	

Attributes

ellipseMode() Changes how ellipse(), circle, and arc() draw.	noSmooth() Disables smoothing when drawing to the canvas.	rectMode() Changes how rect() and square() interpret its parameters.	strokeCap() Sets the shape of the ends of lines.	strokeJoin() Sets the shape of the joins and the shape of the corners for rectangles.
strokeWeight() Sets the thickness of lines.				

Curves

bezier() Draws a bézier curve.	bezierDetail() Sets the number of line segments used to draw bézier curves.	bezierPoint() Calculates a bézier curve coordinate for a given point.	bezierTangent() Calculates the tangent of a point on a bézier curve.	curve() Draws a curve using a Catmull-Rom spline.
curveDetail() Sets the number of line segments used to draw curves.	curvePoint() Calculates a curve coordinate for a given point.	curveTangent() Calculates the tangent of a point on a curve.	curveTightness() Sets the amount the curve bends when it goes through its control points.	

Vertex

beginContour() Begins adding a new hole within a shape.	beginShape() Begins adding vertices to a custom shape.	bezierVertex() Adds a bézier curve segment to a custom shape.	curveVertex() Adds a curve segment to a custom shape.	quadraticVertex() Adds a quadratic bézier curve segment to a custom shape.
vertex() Adds a custom vertex to a custom shape.				

p5.Geometry

calculateBoundings() Calculates the x, y, z coordinates for the boundaries of a 3d geometry.	clearColors() Sets all colors of a 3d geometry to transparent.	computeFaces() Computes faces for a 3d geometry.	computeNormals() Computes vertex normals for a 3d geometry.	cone() Draws a cone.
cylinder() Draws a cylinder.	ellipsoid() Draws an ellipsoid.	extrude() Extrudes a 2d shape.	lathe() Rotates a 2d shape around the x-axis to form a 3d shape.	plane() Draws a plane.
sphere() Draws a sphere.	torus() Draws a torus.			

Color

Creating & Reading

alpha() Gets the alpha (transparency) value of a color.	blue() Gets the blue value of a color.	brightness() Gets the brightness value of a color.	color() Creates a p5.Color object.	green() Gets the green value of a color.
hue() Gets the hue value of a color.	lerp() Calculates a color between two colors at a specified amount.	lerpColor() Calculates a color between two colors at a specified amount.	lightness() Gets the lightness value of a color.	paletteLerp() Generates a lerp color from a color palette.
red() Gets the red value of a color.	saturation() Gets the saturation value of a color.			

Setting

background() Sets the color used for the background of the canvas.	colorMode() Sets the way to parse color values.	fill() Sets the color used to fill shapes.	noFill() Disables filling the shape's interior.	noStroke() Disables drawing the shape's outline.
stroke() Sets the color used to draw lines and borders around shapes.	erase() Begins erasing shapes. Anything drawn after erase() will subtract from what's already drawn.	noErase() Ends erasing. Anything drawn after noErase() will not subtract from what's already drawn.	clear() Clears all pixels from the canvas.	colorMode() Sets the way to parse color values.
fill() Sets the color used to fill shapes.	noFill() Disables filling the shape's interior.	noStroke() Disables drawing the shape's outline.	stroke() Sets the color used to draw lines.	erase() Erases subsequent drawing within a shape defined by beginShape().

p5.Color

setAlpha() Sets the alpha (transparency) value of a color.	setBlue() Sets the blue component of a color.	setGreen() Sets the green component of a color.	setHue() Sets the hue component of a color.	setRed() Sets the red component of a color.
setHue() Sets the hue component of a color.				

Typography

Attributes

textAlign() Sets the way text is aligned when drawing to the canvas.	textAscent() Calculates the maximum amount above the baseline a font can reach.	textDescent() Calculates the maximum amount below the baseline a font can reach.	textLeading() Sets the spacing between text in multiple lines.	textSize() Sets the size of the font used to draw text.
textWidth() Calculates the total width of a string of text when drawn with the current settings.	textWrap() Sets the way to break lines of text when they reach the edge of the canvas.			

Loading & Displaying

loadFont() Loads a font and creates a p5.Font object.	text() Draws text to the canvas.	textFont() Sets the font used by the text() function.	

p5.Font

font Gets the underlying opentype.js font object.	textBounds() Calculates the tight bounding box for a string of text when drawing it.	textToPoints() Computes a set of points that form the outlines of text when drawing it.

Image

createImage() Creates a new p5.Image object.	asArrayBuffer() Saves the current canvas as an image.	asDataURL() Saves the current canvas as an image.	saveCanvas() Saves a canvas or a subset of it as an image file.	saveFrames() Saves a series of frames as image files.
loadImage() Loads an image from a file or URL.				

Loading & Displaying

image() Draws an image to the canvas.	imageMode() Sets how image() interprets its parameters.	loadImage() Loads an image from a file or URL.	noImage() Creates a blank p5.Image object.	nofilter() Removes the current filter for the canvas.
saveGif() Saves an animated GIF from a sketch and saves it to the computer.	tint() Tints images using a color.			

Pixels

blend() Copies a region of pixels from one image to another.	copy() Copies a region of one canvas image to a region of the canvas.	filter() Applies a filter to the canvas.	get() Gets a copy of pixels from the canvas.	loadPixels() Loads the canvas's pixel data into the pixels[] array.
loadPixels() Loads the canvas's pixel data into the pixels[] array.	pixels Stores the canvas's pixel data.	set() Sets the color of a pixel or draws an image to the canvas.	updatePixels() Updates the canvas with the RGBA values in the pixels[] array.	

p5.Image

blend() Copies a region of pixels from one image to another.	copy() Copies a region of one canvas image to the canvas.	delay() Changes the delay between frames in an animated GIF.	filter() Applies an image filter to the canvas.	loadPixels() Loads the image's pixel data into the pixels[] array.
get() Gets a copy of a region of pixels from an image.	getCurrentFrame() Gets the index of the current frame in an animated GIF.	height The image's height in pixels.	loop() Resumes playing a paused GIF image.	mask() Masks part of an image with another image.
pixels An array containing the color of each pixel in the image.	reset() Resets a GIF image to its original state.	resize() Resizes the image to a given width and height.	save() Saves the image to a file.	updatePixels() Updates the image with the RGBA values in the pixels[] array.
width The image's width in pixels.				

Transform

applyMatrix() Applies a custom transformation matrix to the coordinate system.	resetMatrix() Clears all transformations applied to the coordinate system.	rotate() Rotates the coordinate system.	rotateX() Rotates the coordinate system about the x-axis.	rotateZ() Rotates the coordinate system about the z-axis.
scale() Scales the coordinate system.	shearX() Shears the x-axis.	shearY() Shears the y-axis.	translate() Translates the coordinate system.	

Environment

cursor() Changes the mouse appearance.	noCursor() Hides the mouse cursor.	frameRate() Sets the number of frames per second.	getTargetFrameRate() Gets the target frame rate.	noLoop() Stops the draw() function from looping.
loop() Resumes looping after noLoop() has been called.	pixelDensity() Sets the number of pixels per unit of screen or printer.	displayDensity() A function that returns the screen density.	displayHeight() Stores the height of the screen.	fullscreen() Sets or gets the fullscreen mode.
focused Stores whether the sketch is in focus.	frameCount Stores the number of frames drawn since the sketch started.	getItem() Retrieves the value of a key in local storage.	getURL() Returns the current URL, parameters, and hash.	getURLParams() Returns the current URL's parameters as an object.
getURLPath() Returns the pathnames of the current URL as an array.	height A number that stores the window height in pixels.	noSmooth() Disables smoothing when drawing to the canvas.	pixelDensity() Sets the number of pixels per unit of screen or printer.	displayWidth A number that stores the width of the screen.
print() Prints to the console.	storeItem() Stores a key-value pair in local storage.	width A number that stores the window width in pixels.	windowHeight A number that stores the height of the browser's viewport.	windowWidth A number that stores the width of the browser's viewport.

3D

Camera

camera() Sets the position and orientation of the camera.	createCamera() Creates a new camera object.	frustum() Sets the frustum of the camera.	perspective() Sets the perspective projection for the camera.	ortho() Sets an orthographic projection for the camera in a 3D sketch.
setCamera() Sets the camera to a previous one.				

Interaction

debugMode() Adds a grid and an axes to the sketch to visualize the 3D space.	noDebugMode() Turns off debugging for a 3D sketch.	orbitControl() Allows the user to orbit around 3D space with a mouse or touch screen.	resetShader() Resets the current fill and stroke shaders to the default.

Lights

ambientLight() Creates a flat lighting that illuminates all objects equally.	directionalLight() Creates a light that shines in one direction.	imageLight() Uses a 360° image to light the scene.	lightFalloff() Sets how light fades with distance.	lights() Sets default camera and light values for new sketches.
noLights() Disables all lighting.	pointLight() Creates a light that shines from a point in all directions.	setAmbientLight() Sets the color of ambient light.	setDirectionalLight() Sets the color and direction of a directional light.	setPointLight() Sets the color and position of a point light.
spotLight() Creates a light that shines from a point in a cone.	texture() Uses an image to light the shapes.	noStroke() Disables drawing the stroke (outline).	fill() Sets the color used to fill shapes.	

Material

ambientMaterial() Applies ambient lighting to all following surfaces.	specularMaterial() Applies specular lighting to all following surfaces.	shininess() Sets the amount of gloss.	emissiveMaterial() Makes shapes appear to emit light.	shader() Applies a shader to all following surfaces.
resetShader() Resets the current fill and stroke shaders to the default.	createFilterShader() Creates a shader from a built-in p5 filter.	createShader() Creates a new shader object.	loadShader() Loads a shader from files.	noLights() Disables all lighting.
loadModel() Loads a 3D model from a file.	model() Draws a 3D model.	myShader() Stores the default shader used for drawing.	specularColor() Sets the specular color of shapes' surfaces.	texture() Uses an image as the surface texture.
textureWrap() Controls how textures behave when their u or v properties go beyond the edges.				

p5.Camera

camera() Sets the position and orientation of the camera.	centerX The x coordinate of the point where the camera faces.	centerY The y coordinate of the point where the camera faces.	centerZ The z coordinate of the point where the camera faces.	eyeX The x coordinate of the camera.
eyeY The y coordinate of the camera.	eyeZ The z coordinate of the camera.	frustum() Sets the camera's frustum.	lookAt() Points the camera's center at a specified point.	lookCamera() Sets the camera's full state.
move() Moves the camera by a specified amount.	pan() Rotates the camera left and right.	perspective() Sets the camera's perspective.	setCamera() Sets the camera to a previous one.	tilt() Rotates the camera up and down.

p5.Shader

copyToContext() Copies the shader from one drawing context to another.	⚠ loadFrag() Deprecated: Use loadShader() instead. Loads a fragment shader.	setUniform() Sets a shader's uniform (global) variable to a value.

Rendering

blendMode() Sets how pixels are drawn when objects overlap.	clearDepth() Sets the depth buffer to a specified value.	createGraphics() Creates an off-screen p5.Graphics object.	createShader() Creates a new shader object.	createStyle() Creates a new p5.RenderStyle object.
drawingContext() Creates a p5.RenderStyle object.	noClip() Disables clipping.	pixelDensity() Sets the number of pixels per unit of screen or printer.	redraw() Redraws the latest canvas.	resizeCanvas() Resizes the canvas to a given width and height.
setAttributes() Set attributes for the WebGL canvas.				

p5.Framebuffer

autoSized() Toggles the framebuffer's automatic resize when the canvas changes.	begin() Begins drawing objects to the framebuffer.	end() Finishes drawing objects to the framebuffer.	get() Gets a copy of pixels from the framebuffer.	createCamera() Creates a new p5.Camera object to use with the framebuffer.
depth() Gives the framebuffer a depth buffer.	draw() Draws the framebuffer by calling a function that draws to it.	pixelDensity() Sets the framebuffer's pixel density.	pixels An array containing the color of each pixel in the framebuffer.	remove() Deletes the framebuffer from WEBGL memory.
resize() Resizes the framebuffer to a given width and height.	updateSize() Updates the size of the WEBGL canvas to the framebuffer's size.			

p5.Graphics

createFramebuffer() Creates a new p5.Framebuffer object with the same WEBGL context as the parent sketch.	remove() Removes the graphics buffer from the web page.	pixelDensity() Sets the number of pixels per unit of screen or printer.

p5.Renderer

Math

Calculation

abs() Calculates the absolute value of a number.	ceil() Calculates a number rounded up to the nearest integer.	dist() Calculates the distance between two points.	exp() Calculates e^x.	floor() Calculates a number rounded down to the nearest integer.
fract() Calculates the fractional part of a number.	lerp() Calculates a number between two numbers at a specified amount.	log() Calculates the natural logarithm (base e) of a number.	mag() Calculates the magnitude (or length) of a vector.	map() Maps a number from one range to another.
max() Calculates the maximum of two numbers.	min() Calculates the minimum of two numbers.	norm() Normalizes a number from another range into a range of 0 to 1.	pow() Calculates a number raised to a power.	round() Calculates the integer closest to a number.
sq() Calculates the square of a number.	sqrt() Calculates the square root of a number.			

Noise

noise() Returns a Perlin noise value based on the coordinates.	noiseDetail() Adjusts the character and quality of noise.	noiseSeed() Sets a random seed for noise.

Random

random() Returns a random number or a random element from an array.	randomGaussian() Returns a random number fitting a Gaussian, or normal, distribution.	randomSeed() Sets the seed value for the pseudo-random generator.	randomWord() Returns a random word.	randomWords() Returns a random phrase.

Trigonometry

acos() Calculates the arccosine of a number.	asin() Calculates the arcsine of a number.	atan() Calculates the arctangent of a number.	atan2() Calculates the angle formed by a point, the origin, and the x-axis.	cos() Calculates the cosine of an angle.
sin() Calculates the sine of an angle.	tan() Calculates the tangent of an angle.	degrees() Converts radians to degrees.	radians() Converts degrees to radians.	angleMode() Sets the angle mode to radians or degrees.

Vector

createVector() Creates a new p5.Vector object.				

p5.Vector

add() Adds two vectors.	angleBetween() Calculates the angle between two vectors.	array() Converts a vector to an array.	clampToZero() Sets components close to zero to exactly zero.	cross() Calculates the cross product of two vectors.
dot() Calculates the dot product of two vectors.	equals() Checks if two vectors are equal.	fromAngle() Creates a vector from an angle.	fromAngles() Creates a vector from two angles.	fromRandom() Creates a random vector.
get() Gets a copy of a vector.	heading() Calculates the angle of a 2D vector.	lerp() Calculates a vector between two vectors at a specified amount.	mag() Calculates the magnitude (or length) of a vector.	magSq() Calculates the squared magnitude of a vector.
mult() Multiplies a vector by a number or another vector.	normalize() Calculates the unit vector (vector with length of 1).	rem() Calculates the remainder of two vectors.	reflect() Reflects a vector about a line.	rotate() Rotates a 2D vector.
sub() Subtracts one vector from another.				

IO

createInput() Creates a text input.	createButton() Creates a button.	createSlider() Creates a slider.	createCheckbox() Creates a checkbox.	select() Selects an element.
selectAll() Selects all elements matching a CSS selector.	removeElements() Removes all elements created by p5.			

Events

keyIsPressed Stores whether a key is pressed.	keyPressed() Runs once when a key is pressed.	keyReleased() Runs once when a key is released.	keyTyped() Runs when a key is typed.	keyIsDown() Checks if a key is currently pressed.
key Stores the last key pressed.	keyCode Stores the last key code.	mouseX Stores the mouse's x-coordinate.	mouseY Stores the mouse's y-coordinate.	pmouseX Stores the previous mouse x-coordinate.
pmouseY Stores the previous mouse y-coordinate.	mouseIsPressed Stores whether a mouse button is pressed.	mouseButton Stores which mouse button was last pressed.	mouseMoved() Runs once when the mouse moves.	mouseDragged() Runs once when the mouse moves while a button is pressed.
mousePressed() Runs once when a mouse button is pressed.	mouseReleased() Runs once when a mouse button is released.	mouseClicked() Runs once after a mouse button is pressed and released.	doubleClicked() Runs once when the mouse is double-clicked.	mouseWheel() Runs once when the mouse wheel is scrolled.
requestPointerLock() Hides the mouse and locks its position.	exitPointerLock() Exits pointer lock.	touches[] Stores an array of touch points.	touchStarted() Runs once when a touch begins.	touchMoved() Runs once when a touch moves.
touchEnded() Runs once when a touch ends.	deviceOrientation Stores the device's orientation.	deviceMoved() Runs once when the device moves.	deviceTurned() Runs once when the device turns.	deviceShaken() Runs once when the device shakes.
keyPressed(event) Runs once when a key is pressed.	keyReleased(event) Runs once when a key is released.	keyTyped(event) Runs when a key is typed.	mousePressed(event) Runs once when a mouse button is pressed.	mouseReleased(event) Runs once when a mouse button is released.
mouseClicked(event) Runs once after a mouse button is pressed and released.	doubleClicked(event) Runs once when the mouse is double-clicked.	mouseWheel(event) Runs once when the mouse wheel is scrolled.	touchStarted(event) Runs once when a touch begins.	touchMoved(event) Runs once when a touch moves.
touchEnded(event) Runs once when a touch ends.				

DOM

select() Selects an element.	selectAll() Selects all elements matching a CSS selector.	removeElements() Removes all elements created by p5.	createDiv() Creates a div element.	createP() Creates a paragraph element.
createSpan() Creates a span element.	createImg() Creates an image element.	createA() Creates a hyperlink.	createSelect() Creates a dropdown menu.	createRadio() Creates a group of radio buttons.
createColorPicker() Creates a color picker.	createInput() Creates a text input.	createButton() Creates a button.	createCheckbox() Creates a checkbox.	createSlider() Creates a slider.
createVideo() Creates a video element.	createAudio() Creates an audio element.	createCapture() Creates a video capture.	createElement() Creates an HTML element.	

Manipulation

child() Adds an element as a child.	center() Centers an element.	html() Sets the HTML inside an element.	position() Sets the position.	style() Sets a CSS property.
attribute() Sets an attribute.	removeAttribute() Removes an attribute.	addClass() Adds a CSS class.	removeClass() Removes a CSS class.	toggleClass() Toggles a CSS class.
show() Shows an element.	hide() Hides an element.	size() Sets the size.	value() Sets or gets the value.		

Input/Output

changed() Runs when the value changes.	input() Runs when the value changes.	createFileInput() Creates a file input.	createCheckbox() Creates a checkbox.	createSelect() Creates a dropdown menu.
createRadio() Creates a group of radio buttons.	createColorPicker() Creates a color picker.	createButton() Creates a button.	createSlider() Creates a slider.	createInput() Creates a text input.
createVideo() Creates a video element.	createAudio() Creates an audio element.	createCapture() Creates a video capture.	createElement() Creates an HTML element.	

Time & Date

hour() Gets the current hour.	minute() Gets the current minute.	second() Gets the current second.	day() Gets the current day.	month() Gets the current month.
year() Gets the current year.				

`hdot(x, y)` Calculates the dot product of two vectors.	`hmag(x)` Calculates the length/magnitude of a vector.	`hproj(x, y)` Calculates the orthogonal projection of vector x onto vector y.	`hset(x, y, z)` Sets the vector's components to x, y, and z.	`hsub(x, y)` Subtracts the components of y from those of x.	`hzero()` Sets the vector's components to zero.
`hdot(x, y)` Calculates the dot product of two vectors.	`hmag(x)` Calculates the length/magnitude of a vector.	`hproj(x, y)` Calculates the orthogonal projection of vector x onto vector y.	`hset(x, y, z)` Sets the vector's components to x, y, and z.	`hsub(x, y)` Subtracts the components of y from those of x.	`hzero()` Sets the vector's components to zero.
`hnormalize(x)` Normalizes the vector x to a unit length.	`hpolar(x)` Converts a vector's components into polar coordinates.	`hset(x, y, z)` Sets the vector's components to x, y, and z.	`hsub(x, y)` Subtracts the components of y from those of x.	`hzero()` Sets the vector's components to zero.